THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The exterior angle at D of cyclic quadrilateral $ABCD$ (side CD produced) equals the interior angle at B, which is 98°. The interior angle $\angle ADC$ is: 1

(a) 102° **(b)** 92° **(c)** 72° **(d)** 82°

2. AB is a diameter, C and D are on the circle (same side of AB) and $ABCD$ is cyclic with $\angle DAB = 66^\circ$. Then $\angle BCD =$ 1

(a) 114° **(b)** 134° **(c)** 124° **(d)** 104°

3. A chord AB of length 20 cm lies 24 cm from the centre O. The area of triangle OAB (cm^2) is: 1

(a) 242 **(b)** 239 **(c)** 240 **(d)** 241

4. A rhombus inscribed in a circle is a: 1

(a) rectangle that is not a square **(b)** square
(c) kite **(d)** parallelogram only

5. Parallel chords of lengths 6 cm and 8 cm lie on opposite sides of the centre of a circle of radius 5 cm. The distance between them (in cm) is: 1

(a) 6 **(b)** 9 **(c)** 7 **(d)** 8

6. Parallel chords of lengths 6 cm and 8 cm lie on the same side of the centre of a circle of radius 5 cm. The distance between them (in cm) is: 1

(a) 2 **(b)** 0 **(c)** 1 **(d)** 3

7. Two circles of radii 25 cm and 30 cm intersect in a common chord of length 48 cm and their centres lie on opposite sides of it. The distance between the centres (in cm) is: 1

(a) 27 **(b)** 24 **(c)** 25 **(d)** 26

8. $ABCD$ is cyclic, $\angle ABD = 51^\circ$ and $\angle DBC = 29^\circ$. Then $\angle ABC =$ 1

(a) 80° **(b)** 90° **(c)** 100° **(d)** 70°

9. **Assertion (A):** If the angle subtended by an arc at the centre is 100°, the angle it subtends at a point on the remaining part of the circle is 50°. 1

Reason (R): The angle subtended by an arc at the centre is double the angle it subtends at any point on the remaining part of the circle.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Angles in the same segment of a circle are equal. 1
Reason (R): An angle in a semicircle is a right angle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. $ABCD$ is a cyclic quadrilateral and $\angle ABD = 36^\circ$, $\angle DBC = 38^\circ$. Find $\angle ABC$ and $\angle ADC$. 2

12. Two parallel chords of lengths 6 cm and 8 cm lie on the opposite sides of the centre of a circle of radius 5 cm. Find the distance between them. 2

13. Two circles of radii 15 cm and 20 cm intersect, and the common chord is 24 cm. Find the distance between the centres if they lie on opposite sides of the chord. 2

14. $ABCD$ is a cyclic quadrilateral and $\angle ABD = 45^\circ$, $\angle DBC = 22^\circ$. Find $\angle ABC$ and $\angle ADC$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two circles of radii 10 cm and 17 cm intersect in a common chord of length 16 cm. Find the distance between their centres if they are on 3
(i) opposite sides,
(ii) the same side of the common chord.

16. Parallel chords of lengths 12 cm and 16 cm in a circle of radius 10 cm. Find the distance between them in both possible positions. 3

17. AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 68^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. 3

OR

Two circles of radii 17 cm and 25 cm intersect in a common chord of length 30 cm. Find the distance between their centres if they are on 3
(i) opposite sides,
(ii) the same side of the common chord.

18. Parallel chords of lengths 6 cm and 8 cm in a circle of radius 5 cm. Find the distance between them in both possible positions. 3

OR

AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 76^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A kite maker draws a cyclic quadrilateral ABCD on a circular board, with $\angle B = 95^\circ$. The angles $\angle A = (2x + 10)^\circ$ and $\angle C = (3x - 10)^\circ$ are opposite angles. 5
(i) Find $\angle D$.
(ii) Find the value of x.
(iii) Find $\angle A$ and $\angle C$.
(iv) The side BC is extended to E. Find the exterior angle $\angle DCE$ and show that it equals $\angle A$.

20. In a cyclic quadrilateral $ABCD$, $\angle A = (x + 30)^\circ$, $\angle B = (3y + 20)^\circ$, $\angle C = (2x + 15)^\circ$ and $\angle D = (2y + 5)^\circ$. Find x, y and all four angles. 5

OR

A circle of radius 26 cm has a chord AB of length 20 cm. Find the distance of AB from the centre O, the area of $\triangle OAB$ and the perimeter of $\triangle OAB$. 5

END OF QUESTION PAPER — **Total: 40 Marks**